

Notes: Custódio and Metzger (JFE, 2014)

Financial Expert CEOs: CEO's Work Experience and Firm's Financial Policies

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1 Big picture

The paper studies whether a CEO’s prior career experience in finance affects corporate financial policy. The main empirical object is a **financial expert CEO**: a CEO with prior work experience in banking or investment firms, in finance-related corporate roles such as CFO or treasurer, or in a major auditing firm.

The core finding is that firms led by financial expert CEOs look financially more aggressive and financially more flexible. They hold less cash, use more debt, and repurchase shares more often. The paper then asks whether these patterns reflect financial skill, easier access to external finance, or simply endogenous matching between mature firms and finance-oriented CEOs.

The authors’ answer is nuanced. The evidence is broadly consistent with a **financial skills view**: financial expert CEOs seem more sophisticated in applying finance theory, react more actively to credit and tax shocks, and reduce the firm’s reliance on internal cash flows. At the same time, the paper is careful that CEO-firm matching cannot be fully ruled out: financial expert CEOs are more likely to be appointed by mature firms.

2 Incremental contribution of the paper

This paper contributes incrementally in five ways.

- **CEO background rather than only CEO traits.** Earlier work often studies CEO age, MBA education, overconfidence, or fixed effects. This paper focuses on the CEO’s employment history and asks whether finance-related work experience matters for corporate policy.
- **Micro evidence on the mechanism.** Instead of only showing that financial expert CEOs are correlated with lower cash or higher leverage, the paper studies WACC-fallacy behavior, analyst forecast dispersion, investment-cash flow sensitivity, credit shocks, tax shocks, CFO turnover, and lending relationships.
- **A distinction between expertise and conflicted finance ties.** In contrast to board bankers who may benefit their banks, financial expert CEOs are internal decision makers. The paper finds less investment-cash flow sensitivity but no clear evidence of overinvestment.
- **Direct discussion of CEO-firm matching.** The paper explicitly studies whether mature firms are more likely to hire financial expert CEOs and then uses life-cycle controls, retirement turnover tests, and matching estimators to check robustness.
- **Costs of financial expertise.** The analysis extends beyond finance and finds evidence that financial expert CEOs are associated with less R&D and weaker innovation output, suggesting that finance expertise may come with trade-offs.

3 Data and measurement

3.1 Sample construction

The initial panel comes from ExecuComp and contains CEO-firm observations for S&P 1500 firms. The authors match CEOs to BoardEx to obtain biographical information and past employment

histories. They exclude financial firms and utilities because those sectors have special capital structure and regulatory constraints.

The final main sample covers 1993–2007 and contains 17,716 CEO-firm-year observations and 4,277 unique CEOs. Firm accounting data come from Compustat, returns from CRSP, analyst forecasts from I/B/E/S, and loan facilities from Dealscan.

3.2 Financial expert CEO

A CEO is classified as a financial expert if the CEO has past experience in at least one of the following:

- a banking, nondepository financial, investment banking, or securities firm;
- a finance-related corporate role such as accountant, CFO, treasurer, or VP of finance;
- a major auditing firm.

About 41% of the CEO-firm-year observations have financial expert CEOs. The classification is intentionally based on career experience, not only on formal finance education.

3.3 Main outcomes

The paper studies three groups of outcomes.

- **Financial policy:** cash holdings, book leverage, dividend payout, share repurchases, debt issuance, and total payout.
- **Mechanisms:** WACC-fallacy investment distortions, analyst forecast dispersion, investment-cash flow sensitivity, credit-market response, tax-cut response, CFO turnover, and lender switching.
- **Nonfinancial policy:** R&D, patents, citations, advertising, sales sensitivity to advertising, and average wages.

4 Empirical framework

The baseline empirical specification relates a firm policy outcome to a financial expert CEO indicator and controls:

$$Y_{it} = \alpha + \beta \text{FinancialExpertCEO}_{it} + X'_{it}\gamma + \text{CEO}'_{it}\delta + FE + \varepsilon_{it}. \quad (1)$$

Depending on the outcome, the paper uses industry-year fixed effects or firm fixed effects. The firm fixed effects specifications identify effects from within-firm CEO switches from nonfinancial expert to financial expert CEOs or vice versa.

For investment-cash flow sensitivity, the key specification is:

$$\text{Capex}_{it} = \alpha + \beta \text{CashFlow}_{it} + \theta(\text{FinancialExpertCEO}_{it} \times \text{CashFlow}_{it}) + X'_{it}\gamma + FE + \varepsilon_{it}. \quad (2)$$

A negative θ means investment is less dependent on internal funds when the CEO has financial expertise.

For shock-based tests, the paper uses interactions with aggregate credit conditions and the 2003 dividend and capital gains tax cuts:

$$Y_{it} = \alpha + \beta(\text{FinancialExpertCEO}_{it} \times \text{DefaultSpread}_t) + X'_{it}\gamma + FE + \varepsilon_{it}, \quad (3)$$

and

$$\text{Payout}_{it} = \alpha + \beta(\text{FinancialExpertCEO}_{it} \times \text{Post2003}_t) + X'_{it}\gamma + FE + \varepsilon_{it}. \quad (4)$$

5 Identification and interpretation

The main identification challenge is **CEO-firm matching**. Financial expert CEOs may not randomly arrive at firms; they may be selected by firms that already want lower cash, higher leverage, or more payout.

The paper uses several approaches to address this concern:

- firm fixed effects, which absorb time-invariant firm heterogeneity;
- controls for CEO education, generalist ability, age, tenure, and other CEO traits;
- direct tests of matching, showing that financial expert CEOs are more likely to be hired by mature firms;
- life-cycle controls and industry-by-year fixed effects;
- likely exogenous turnover timing after retiring CEOs;
- matching estimators that pair firms with and without financial expert CEOs on observables.

The authors do not claim a perfect natural experiment. The contribution is a coherent set of results consistent with financial expertise mattering, while acknowledging that endogenous matching may explain part of the patterns.

6 Main findings

6.1 Financial expert CEOs and financial policy

Financial expert CEOs are associated with:

- about 12% lower cash holdings;
- about 6% higher leverage for the average firm;
- a higher propensity to repurchase shares;
- little robust difference in dividend initiation once firm fixed effects are included.

The first-order message is that financial expert CEOs run firms with lower precautionary liquidity and greater use of external financing.

6.2 Financial sophistication

The paper gives several pieces of mechanism evidence.

- In diversified firms, nonfinancial expert CEOs display investment distortions consistent with the WACC fallacy, while financial expert CEOs do not.
- Analyst forecast dispersion is lower for firms with financial expert CEOs in cross-sectional specifications, suggesting better communication with capital markets, although this result is not robust to firm fixed effects.
- Investment is less sensitive to cash flow when the CEO is a financial expert, which is consistent with easier access to outside capital.
- Financial expert CEOs increase cash and debt issuance more when default spreads are high.
- Financial expert CEOs increase total payout more after the 2003 dividend and capital gains tax cuts.

6.3 Matching and trade-offs

Financial expert CEOs are more likely to be hired by larger, older, more mature firms. Adding life-cycle controls and using retirement-turnover subsamples does not remove the main cash, leverage, and payout patterns.

The paper also finds that firms led by financial expert CEOs spend less on R&D and generate fewer patents and citations. This suggests that financial expertise may help implement financial policy but may not be a free lunch for innovation-intensive firms.

7 Tables and figures

7.1 Variables and summary statistics

Table 1 Variable definitions.	
Panel A: CEO characteristics	
Variable	Definition
Board independence	Ratio of number of independent directors to board size.
CEO age	Age of CEO in years.
CEO tenure	Number of years as CEO in the current position.
CEO-chair dummy	Dummy variable equal to one if the CEO is also chairman of the board.
Compensation equity	Equity pay - Value of restricted stock granted plus value of options granted in thousands of dollars (Edecump RSTGRNT + OPTION_AWARDS_BK.VALUEEdecump).
Compensation total	Total CEO pay in thousands of dollars, which consists of salary, bonus, value of restricted stock granted, value of options granted, long-term incentive payout, and other compensation (Edecump TDC1). (Edecump).
Econ dummy	Dummy variable that takes the value of 1 if CEO has an economics related degree, and zero otherwise.
External hire dummy	Dummy variable equal to one if the CEO is hired externally; dummy equal to zero if the CEO is internally appointed.
Fast track CEO	Age at which CEO became CEO for the first time.
Financial expert CEO	Number of firms with past experience in either banking or investment firms (two-digit SIC codes 60, 61, and 62); in a finance-related role (Accountant, CFO, Treasurer, or VP of Finance), or in a large auditing firm (current and former top-tier companies: Pricewaterhouse, Deloitte, Ernst & Young, KPMG, Arthur Andersen, Coopers, PwC, Marwick, Touche Ross).
General Ability Index	First factor of applying principal components analysis to five proxies of general managerial ability: past number of positions, number of firms, number of industries, CEO experience, and conglomerate experience.
Ivy League alumnus	Graduated from an Ivy League college.
Law dummy	Dummy variable that takes the value of one if CEO has a law-related degree, and zero otherwise.
MBA dummy	Dummy variable that takes the value of one if CEO has an MBA degree, and zero otherwise.
Recession graduate	Graduated during NBER recession.
Science dummy	Dummy variable that takes the value of one if CEO has a science-related degree, and zero otherwise.
Sys dummy	A dummy equal to one if the CEO is male.
Panel B: Firm characteristics	
Variable	Definition
Acquisitions	Value of acquisitions (AQ).
Advertising	Advertising expenses (XAD).
Analyst dispersion	Standard deviation across different analyses of yearly earnings forecasts standardized by the assets of the firm (from I/B/E/S).
Assets	Book value of assets (AT).
Asset growth	Book value of assets (AT) in year t over book value of assets in year $t-1$, minus one.
Asset volatility	Standard deviation of stock return during the fiscal year times market value of equity (CSHO * PRCC_F) divided by market value of assets (AT + CSHO*PRCC_F - CEQ).
Book leverage	Ratio of total debt (DXT) to book value of assets (AT).
Capex	Ratio of capital expenditures (CAPX) to book value of assets (AT).
Cash	Ratio of cash and short-term investments (CHE) over book value of total assets (AT).
CFO turnover	A dummy variable that is equal to one if there is a CFO turnover.
Citations	Total number of citations received by the set of patents that a firm applied for in a given year from NBER patent database.
Citations (Q)	Adjusted total number of citations received by the set of patents that a firm applied for in a given year from NBER patent database. The adjustment uses the weighting index from Hall, Jaffe, and Trajtenberg (2001, 2005).
Debt issues	Dummy variable equal to one if the firm issues debt in a given year (from Dealogic).
Diversification dummy	Dummy variable that takes the value of one if a firm has more than one business segment and zero otherwise.
Dividend dummy	Dummy variable that takes the value of one if the firm pays dividends (DVC) and zero otherwise.
Firm age	Number of years between fiscal year (FYEAD) and CSOP listing year (LSTYEAD).
Firm size	Book value of assets (AT).
Firm size	Institutional ownership using data from the Thomson CDA/Spectrum 13 F Holdings (see Aghion, Van Reenen, and Zingales, 2013).
Net working capital	Net current assets minus cash (ACT - CHE - LCT) divided by assets (AT) minus cash (CHE).
Patents	Number of patent applications from NBER patent database.
PPE	Ratio of net property, plant, and equipment (PPNET) to book value of assets (AT).
R&D	Ratio of research and development expenditures (XRD) to book value of assets (AT).
Rating dummy	Dummy variable that takes the value of one if a firm has a S&P domestic long-term issuer credit rating (SPDRCRM).
Repurchase dummy	Dummy variable equal to one if the firm buys back shares in a given year.
Retained earnings	Ratio of retained earnings (RE) and common equity (CEQ).
Return on assets (ROA)	Ratio of earnings before interest, taxes, depreciation and amortization (EBITDA) to book value of total assets (AT).
Sales	Sales (SALE).
Stock returns	Annual stock return $(PRCC_F)_t / (ADJRET) + DVPSSK_F(t) / (ADJRET) / (PRCC_F(t) / (ADJRET) - 1)$.
Switch bank	Dummy variable that is equal to one if the leading loan provider of the first new loan after the CEO turnover did not offer the last loan before the turnover (or had not provided any loan in the last three years before the turnover).
Tobin's q	Ratio of market value of assets (AT + CSHO*PRCC_F - CEQ) to book value of total assets (AT).
Total payout	Dividends plus share repurchases divided by net income.
Wage (average)	Total wage expense divided by the total number of employees in a given year.
Panel C: Market variables	
Variable	Definition
Default spread	Difference between the average yield on Moody's corporate bond ratings AAA and BAA.
Recession dummy	One if a given year is considered to be an NBER recession.

(a) Table 1. Variable definitions

Table 2 The sample consists of ExecuComp firms for which CEO profile data are available from Boardfiles in the 1993-2007 period. All variables are winsorized at the 1st and 99th values. Variable definitions are as defined in Table 1. ***p < 0.01, **p < 0.05, *p < 0.1.											
Panel A: CEO financial expertise											
	Mean	p25	p50	p75	sd	N					
Financial expert dummy	0.413	0.000	0.000	1.000	0.492	17736					
Depository institutions	0.357	0.000	0.000	0.000	0.364	17736					
Nondepository institutions	0.048	0.000	0.000	0.000	0.214	17736					
Investment banks & securities brokerages	0.001	0.000	0.000	0.000	0.020	17736					
Auditing experience	0.074	0.000	0.000	0.000	0.261	17736					
CFO role	0.002	0.000	0.000	0.000	0.023	17736					
Financial role	0.039	0.000	0.000	0.000	0.170	17736					
Panel B: CEO characteristics											
	Mean	p25	p50	p75	sd	N	Financial expert CEO	Nonfinancial expert CEO	DIF	S.E.	
CEO age	51.634	50.000	50.000	50.000	7.773	17698	0.300	0.330**	0.030**	(0.010)	
Sex dummy	0.983	1.000	1.000	1.000	0.130	17736	0.981	0.984	-0.003	(0.002)	
CEO tenure	7.777	2.000	5.000	8.000	7.522	16699	7.105	7.227	-0.122	(0.130)	
MBA dummy	0.371	0.000	0.000	1.000	0.483	17736	0.370	0.371	-0.001	(0.007)	
Law dummy	0.227	0.000	0.000	1.000	0.409	17736	0.238	0.238	-0.011**	(0.007)	
Science dummy	0.303	0.000	0.000	1.000	0.458	17736	0.305	0.305	-0.002	(0.007)	
Low dummy	0.103	0.000	0.000	0.000	0.308	17736	0.098	0.214	-0.114**	(0.006)	
Ivy League alumnus	0.172	0.000	0.000	0.000	0.378	17736	0.177	0.146	0.030**	(0.006)	
Recession graduate	0.286	0.000	0.000	1.000	0.452	15485	0.311	0.265	0.046**	(0.007)	
General ability	-0.009	-0.170	-0.082	0.344	0.875	15485	0.232	-0.094	0.400**	(0.015)	
Fast track CEO	47.077	42.000	46.000	54.000	8.537	14227	48.139	43.824	4.315**	(0.144)	
External hire dummy	0.403	0.000	0.000	1.000	0.490	15485	0.381	0.421	-0.039**	(0.008)	
CEO-chair dummy	0.069	0.000	0.000	1.000	0.490	15485	0.054	0.257	-0.202**	(0.008)	
Total compensation	4523.889	1860.605	2222.096	4266.493	5330.708	15485	4346.638	3795.761	550.877**	(88.153)	
Cash compensation	1173.623	590.732	878.828	1422.075	2066.468	15485	1283.427	988.274	295.153**	(58.816)	
Equity compensation	2350.265	1267.795	1343.268	2844.425	4473.028	13983	2616.260	2210.103	406.157**	(77.272)	
Panel C: Firm characteristics											
	Mean	p25	p50	p75	sd	N	Financial expert CEO	Nonfinancial expert CEO	DIF	S.E.	
Cash	0.267	0.023	0.078	0.275	0.406	17736	0.230	0.354	-0.114**	(0.008)	
Book leverage	0.236	0.040	0.200	0.328	0.182	17736	0.238	0.262	-0.024**	(0.003)	
Dividend dummy	0.481	0.000	0.000	1.000	0.500	17736	0.562	0.424	0.138**	(0.008)	
Share repurchase dummy	0.578	0.000	1.000	1.000	0.494	17736	0.615	0.553	0.062**	(0.008)	
Analyst dispersion	0.000	0.001	0.002	0.006	0.040	9051	0.005	0.012	-0.007**	(0.001)	
Assets	4480.160	447.710	180.703	3188.079	1206.040	17736	3994.446	3335.155	2638.291**	(183.76)	
Tobin's q	2.137	1.303	1.465	2.425	1.443	17736	2.448	2.398	-0.10**	(0.022)	
ROA	0.138	0.092	0.140	0.195	0.305	17736	0.144	0.154	-0.01**	(0.001)	
Asset volatility	0.263	0.003	0.047	0.081	0.055	17736	0.184	0.209	-0.025**	(0.001)	
Firm age	21.001	8.000	15.000	30.000	18.371	17540	24.276	18.863	5.413**	(0.279)	
Capex	0.025	0.000	0.000	0.000	0.055	17736	0.062	0.005	-0.057**	(0.001)	
R&D	0.034	0.000	0.002	0.044	0.058	17736	0.026	0.040	-0.014**	(0.001)	
High-tech dummy	0.205	0.000	0.000	0.000	0.404	17736	0.178	0.220	-0.042**	(0.001)	
Asset growth	0.135	-0.012	0.071	0.391	0.312	15906	0.126	0.142	-0.016**	(0.005)	
Acquisitions	0.015	0.000	0.000	0.012	0.079	17736	0.036	0.035	0.001	(0.001)	
Diversification dummy	0.580	0.000	1.000	1.000	0.494	10550	0.600	0.564	0.036**	(0.01)	
Net working capital	0.000	-0.016	0.001	0.211	0.108	17736	0.077	0.008	-0.069**	(0.003)	
Retained earnings	0.300	0.195	0.570	0.867	1.589	17875	0.452	0.296	0.156**	(0.024)	

(b) Table 2. CEO financial expertise and summary statistics

Read: Table 1 defines the CEO variables, firm policy variables, and market variables used throughout the paper. Table 2 shows that 41% of CEOs are classified as financial experts. Financial expert CEOs differ from nonexperts in education, career breadth, and the types of firms they lead.

Economic message: The empirical design depends critically on distinguishing work experience from other CEO attributes. Table 2 shows why the paper must control for firm maturity and CEO background: financial expert CEOs are not randomly distributed across firms.

7.2 Baseline financial policy and WACC-fallacy mechanism

Table 3

The dependent variable in regressions (1)-(2) is the log of cash and marketable securities. The dependent variable in regressions (3)-(4) is the ratio of total debt to assets. The dependent variable in regressions (5)-(6) is a dummy variable equal to one if the firm paid a dividend in the current year. The dependent variable in regressions (7)-(8) is a dummy variable equal to one if the firm repurchased shares in the current year. Variable definitions are as defined in Table 1. Standard errors clustered at firm-level in parentheses. ***p < 0.01, **p < 0.05, *p < 0.1.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Cash		Leverage		Dividends		Repurchases	
Financial expert CEO	-0.123*** (0.040)	-0.114** (0.051)	0.009* (0.005)	0.013** (0.006)	0.007** (0.003)	0.011 (0.009)	0.021*** (0.007)	0.036** (0.016)
Age	-0.030 (0.022)	-0.050* (0.026)	-0.000 (0.001)	-0.000 (0.001)	0.003 (0.002)	0.003 (0.004)	0.005 (0.005)	0.003 (0.009)
Age squared	0.000 (0.000)	0.000** (0.000)	0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
Sex	-0.336** (0.199)	-0.073 (0.187)	0.018 (0.020)	-0.012 (0.017)	-0.004 (0.013)	-0.040 (0.018)	0.001 (0.027)	-0.026 (0.049)
Tenure	0.002 (0.003)	-0.008** (0.004)	-0.001* (0.000)	-0.000 (0.001)	0.000 (0.000)	0.000 (0.001)	-0.000 (0.001)	0.001 (0.002)
Log(assets)	-0.216*** (0.017)	-0.476*** (0.035)	0.013*** (0.002)	0.018*** (0.006)	0.010*** (0.001)	0.032*** (0.001)	0.027*** (0.003)	0.065*** (0.014)
Log q	0.632*** (0.035)	0.444*** (0.036)	-0.001 (0.007)	-0.008 (0.006)	-0.001 (0.007)	-0.008 (0.006)	-0.001 (0.007)	-0.026 (0.006)
Asset volatility	1.239*** (0.107)	0.412*** (0.087)	-0.323*** (0.020)	-0.155*** (0.015)	-0.155*** (0.015)	-0.155*** (0.015)	-0.155*** (0.015)	-0.155*** (0.015)
R&D	2.636*** (0.207)	0.601*** (0.243)	0.031 (0.036)	-0.055 (0.050)	-0.055 (0.050)	-0.055 (0.050)	-0.055 (0.050)	-0.055 (0.050)
Capex	-4.685*** (0.414)	-3.251*** (0.350)	-0.240*** (0.063)	-0.206*** (0.051)	-0.206*** (0.051)	-0.206*** (0.051)	-0.206*** (0.051)	-0.206*** (0.051)
Dividend dummy	0.143*** (0.048)	0.098* (0.053)	-0.048*** (0.006)	-0.052*** (0.008)	-0.052*** (0.008)	-0.052*** (0.008)	-0.052*** (0.008)	-0.052*** (0.008)
ROA	-0.145*** (0.041)	-0.098*** (0.041)	-0.048*** (0.006)	-0.052*** (0.008)	-0.052*** (0.008)	-0.052*** (0.008)	-0.052*** (0.008)	-0.052*** (0.008)
PPE	0.153*** (0.025)	0.121*** (0.025)	0.153*** (0.018)	0.121*** (0.018)	0.153*** (0.018)	0.121*** (0.018)	0.153*** (0.018)	0.121*** (0.018)
Leverage	0.079 (0.082)	0.305*** (0.079)	-0.055*** (0.011)	-0.085*** (0.025)	-0.085*** (0.025)	-0.085*** (0.025)	-0.085*** (0.025)	-0.085*** (0.025)
Cash flow	1.041*** (0.109)	0.468*** (0.084)	-0.055*** (0.011)	-0.085*** (0.025)	-0.085*** (0.025)	-0.085*** (0.025)	-0.085*** (0.025)	-0.085*** (0.025)
Net working capital	-1.145*** (0.131)	-0.468*** (0.133)	-0.055*** (0.011)	-0.085*** (0.025)	-0.085*** (0.025)	-0.085*** (0.025)	-0.085*** (0.025)	-0.085*** (0.025)
Acquisitions	-2.349*** (0.158)	-1.074*** (0.131)	-0.055*** (0.011)	-0.085*** (0.025)	-0.085*** (0.025)	-0.085*** (0.025)	-0.085*** (0.025)	-0.085*** (0.025)
Dividend dummy(t-1)			0.900*** (0.005)	0.605*** (0.019)	0.900*** (0.005)	0.605*** (0.019)	0.900*** (0.005)	0.605*** (0.019)
Repurchase (t-1)					0.520*** (0.009)	0.251*** (0.013)	0.520*** (0.009)	0.251*** (0.013)
Asset growth			-0.001 (0.005)	-0.003 (0.007)	-0.002*** (0.002)	-0.008*** (0.012)	-0.008*** (0.012)	-0.008*** (0.012)
Cash			0.005 (0.003)	0.029*** (0.008)	0.029*** (0.008)	0.029*** (0.008)	0.029*** (0.008)	0.029*** (0.008)
MBA		0.017 (0.054)	-0.001 (0.007)	-0.004 (0.010)	-0.004 (0.010)	-0.004 (0.010)	-0.004 (0.010)	-0.004 (0.010)
Econ		-0.001 (0.054)	-0.001 (0.007)	-0.010 (0.010)	-0.010 (0.010)	-0.010 (0.010)	-0.010 (0.010)	-0.010 (0.010)
Science		-0.093 (0.059)	0.004 (0.008)	0.000 (0.010)	0.000 (0.010)	0.000 (0.010)	0.000 (0.010)	0.000 (0.010)
Law		-0.011 (0.063)	0.009 (0.008)	-0.023 (0.010)	-0.023 (0.010)	-0.023 (0.010)	-0.023 (0.010)	-0.023 (0.010)
Ivy League alumnus		-0.015 (0.063)	0.008 (0.008)	-0.010 (0.010)	-0.010 (0.010)	-0.010 (0.010)	-0.010 (0.010)	-0.010 (0.010)
Generalist index		0.014 (0.028)	-0.001 (0.003)	-0.004 (0.004)	-0.004 (0.004)	-0.004 (0.004)	-0.004 (0.004)	-0.004 (0.004)
Recession graduate		0.006 (0.050)	-0.001 (0.006)	-0.010 (0.009)	-0.010 (0.009)	-0.010 (0.009)	-0.010 (0.009)	-0.010 (0.009)
Fast track CEO		-0.003 (0.003)	0.000 (0.000)	0.000 (0.001)	0.000 (0.001)	0.000 (0.001)	0.000 (0.001)	0.000 (0.001)
Obs	16191	13504	16275	13667	14138	11986	14138	11986
Industry dummies	Yes	No	Yes	No	Yes	No	Yes	No
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	No	Yes	No	Yes	No	Yes	No	Yes

(a) Table 3. Financial expert CEOs and financial policy

Table 4

This table shows the effect of financial experience on potential overinvestment (underinvestment) in high (low) beta division because of applying a companywide discount rate instead of a division-specific one. We follow Krüger, Lander, and Thesmar (2011), Table 5. "Using business segment data from Compustat, we construct industry-level divisions by aggregating segment. A division is defined as the bundle of arm's segments operating in the same F48 industry. The regressions are run on divisions which do not have the highest sales in the conglomerate (non-core divisions). The dependent variable Capex/Assets is division-level capital expenditures in period $t+1$ scaled by division assets in period t . Delta beta is the difference between the industry-level asset beta of the core-division (i.e. the division with the highest sales) and the industry-level asset beta of the non-core division. Division q is the division's industry-level Tobin's q . Core q is the industry-level Tobin's q of the division with the highest sales in the conglomerate. Both are calculated for a sample of standalone firms from the same industry. Firm cash flow is the firm's cash flow scaled by total assets. Division size is the logarithm of division sales and Firm size is the log of total assets. Firm age is the logarithm of the current year plus one minus the year in which the firm first appeared in the Compustat North America database. CEO and firm variable definitions are as defined in Table 1. Standard errors clustered at firm-level in parentheses. ***p < 0.01, **p < 0.05, *p < 0.1.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Capex(non-core)						
Delta beta	0.020*** (0.006)	0.028*** (0.008)	0.012 (0.008)	0.030*** (0.007)	0.008 (0.008)	0.020*** (0.007)	0.009 (0.009)
Segment q				0.000 (0.000)	-0.000* (0.000)	0.000 (0.000)	-0.000 (0.000)
Core q				-0.000** (0.000)	-0.000 (0.000)	-0.000** (0.000)	0.000 (0.000)
Segment size				-0.003 (0.002)	-0.005*** (0.002)	-0.002 (0.003)	-0.002*** (0.003)
Cash flow				0.025 (0.018)	0.042* (0.023)	0.008 (0.021)	0.035* (0.020)
Firm size				0.000 (0.004)	0.005* (0.003)	-0.002 (0.004)	0.006** (0.003)
Firm age				0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Asset growth						0.009 (0.010)	-0.003 (0.006)
Retained earnings						-0.001 (0.001)	-0.000 (0.001)
Age						-0.001 (0.002)	-0.001 (0.003)
Age squared						0.000 (0.000)	-0.000 (0.000)
Gender						-0.026 (0.024)	0.047* (0.027)
Tenure						0.000 (0.000)	0.000 (0.000)
Econ						0.006 (0.005)	0.011** (0.005)
Science						0.005 (0.005)	0.008 (0.005)
Law						-0.017*** (0.005)	0.004 (0.006)
MBA						0.001 (0.005)	-0.007 (0.005)
Ivy League alumnus						0.005 (0.005)	0.002 (0.005)
Generalist index						-0.000 (0.005)	0.001 (0.005)
Recession graduate						-0.002 (0.002)	-0.002 (0.002)
Fast track CEO						-0.005 (0.005)	-0.004 (0.004)
Sample	All	Nonfinancial experts	Financial experts	Nonfinancial experts	Financial experts	Nonfinancial experts	Financial experts
Obs	4976	2564	2412	2549	2401	1972	1972
Industry dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R-squared	0.102	0.121	0.108	0.130	0.123	0.139	0.122

(b) Table 4. Financial expertise and WACC-fallacy behavior

Read: Table 3 shows that financial expert CEOs are associated with lower cash, higher leverage, and more share repurchases. The cash coefficient is around -0.12 in both OLS and firm fixed effects specifications, and the leverage coefficient is positive. Table 4 shows that the WACC-fallacy pattern is strong for nonfinancial expert CEOs but not for financial expert CEOs.

Economic message: The baseline evidence is not merely that financial expert CEOs choose different policies. The WACC-fallacy test suggests they may make capital allocation decisions in a way that is closer to standard finance theory.

7.3 Communication and investment-cash flow sensitivity

Table 5

This table shows the effect of financial experience on the dispersion of analyst forecasts. The dependent variable in all regressions is standard deviation of yearly earnings forecasts divided by non-cash assets. CEO and firm variable definitions are as defined in Table 1. Standard errors clustered at firm-level in parentheses. ***p < 0.01, **p < 0.05, *p < 0.1.

Panel A: Univariate analysis			
Financial expert CEO	Nonfinancial expert CEO	Diff.	s.e.
0.005	Analyst forecast dispersion 0.012	-0.007***	(0.001)
Panel B: Multivariate analysis			
	(1)	(2)	
Financial expert CEO	-0.002** (0.001)	Analyst forecast dispersion -0.000 (0.000)	
Log(assets)	-0.007*** (0.001)	-0.006*** (0.002)	
Firm age	0.005** (0.000)	0.000 (0.000)	
Tobin's q	0.005 (0.004)	-0.003 (0.002)	
Asset growth	-0.008*** (0.002)	-0.002 (0.002)	
Retained earnings	-0.000 (0.001)	-0.001* (0.000)	
Asset volatility	0.015 (0.020)	0.007 (0.004)	
R&D	0.026 (0.022)	0.043 (0.033)	
Age	0.000 (0.001)	-0.001 (0.001)	
Age squared	-0.000 (0.000)	0.000 (0.000)	
Sex	0.001 (0.004)	0.001 (0.001)	
Tenure	-0.000 (0.000)	0.000 (0.000)	
MBA	-0.000 (0.001)	0.000 (0.001)	
Econ	-0.001 (0.001)	0.000 (0.000)	
Science	-0.001 (0.001)	-0.001 (0.001)	
Law	-0.001 (0.001)	-0.001 (0.001)	
Ivy League alumnus	-0.001 (0.001)	-0.001 (0.001)	
Generalist index	0.000 (0.000)	0.000 (0.000)	
Recession graduate	-0.001 (0.001)	-0.001 (0.001)	
Fast track CEO	0.000 (0.000)	0.000 (0.000)	
Obs	8128	7462	
Industry dummies	Yes	No	
Year dummies	Yes	Yes	
Firm fixed effects	No	Yes	
Adjusted R-squared	0.087	0.529	

From the results in this section we conclude that there is some evidence that financial expert CEOs are better at communicating firm prospects with other entities in

be a consequence of better access to financing by this type of CEO. Table 6 shows the results. We run OLS and firm fixed effects regressions of investment, measured by the

Table 6

The dependent variable in regressions (1) to (4) is the ratio of capital expenditures to assets. Variable definitions are included in Table 1. Standard errors clustered at firm-level in parentheses. ***p < 0.01, **p < 0.05, *p < 0.1.

	(1)	(2)	(3)	(4)
Capex				
Fin. exp. CEO × Cash flow	-0.045*** (0.015)	-0.044*** (0.015)	-0.056*** (0.015)	-0.050*** (0.015)
Financial expert CEO	0.001 (0.002)	0.005** (0.002)	0.002 (0.002)	0.006** (0.002)
Tobin's q (t-1)	0.001 (0.000)	0.003*** (0.001)	0.000 (0.000)	0.002*** (0.000)
Log assets (t-1)	-0.003*** (0.001)	-0.001*** (0.001)	-0.003*** (0.001)	-0.001*** (0.001)
Log assets × Cash flow (t-1)	0.012*** (0.001)	0.012*** (0.001)	0.012*** (0.001)	0.012*** (0.001)
Board size	-0.009* (0.005)	-0.006 (0.005)	-0.004 (0.005)	-0.002 (0.005)
Board size × Cash flow	0.064 (0.040)	0.015 (0.034)	0.033 (0.037)	0.008 (0.035)
Board independence	-0.003 (0.006)	-0.007 (0.007)	0.000 (0.006)	-0.010 (0.007)
Board independence × Cash flow	0.060 (0.049)	0.069 (0.052)	0.058 (0.054)	0.087 (0.056)
Age	-0.002** (0.001)	-0.002** (0.001)	-0.001 (0.001)	-0.001 (0.001)
Age × Cash flow	0.005 (0.011)	0.005 (0.011)	0.001 (0.010)	0.001 (0.010)
Age squared	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Age squared × Cash flow	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
Sex	0.008 (0.005)	0.008 (0.005)	-0.005 (0.005)	-0.005 (0.005)
Sex × Cash flow	-0.026 (0.044)	-0.026 (0.044)	-0.005 (0.026)	-0.005 (0.026)
Tenure	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Tenure × Cash flow	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)
MBA	-0.002 (0.002)	-0.002 (0.002)	0.001 (0.002)	0.001 (0.002)
MBA × Cash flow	0.017 (0.015)	0.017 (0.015)	-0.004 (0.015)	-0.004 (0.015)
Econ	0.001 (0.002)	0.001 (0.002)	-0.001 (0.003)	-0.001 (0.003)
Econ × Cash flow	0.007 (0.010)	0.007 (0.010)	0.011 (0.020)	0.011 (0.020)
Science	0.001 (0.002)	0.001 (0.002)	-0.000 (0.003)	-0.000 (0.003)
Science × Cash flow	0.008 (0.010)	0.008 (0.010)	-0.003 (0.021)	-0.003 (0.021)
Law	-0.008** (0.004)	-0.008** (0.004)	-0.007* (0.004)	-0.007* (0.004)
Law × Cash flow	0.028 (0.032)	0.028 (0.032)	0.011 (0.027)	0.011 (0.027)
Ivy League alumnus	0.003 (0.002)	0.003 (0.002)	0.002 (0.003)	0.002 (0.003)
Ivy League alumnus × Cash flow	-0.006 (0.016)	-0.006 (0.016)	-0.019 (0.019)	-0.019 (0.019)
Generalist index	-0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)
Generalist × Cash flow	0.002 (0.009)	0.002 (0.009)	0.002 (0.010)	0.002 (0.010)
Recession graduate	-0.000 (0.002)	-0.000 (0.002)	-0.000 (0.003)	-0.000 (0.003)
Recession graduate × Cash flow	0.014 (0.017)	0.014 (0.017)	-0.012 (0.017)	-0.012 (0.017)
Obs	10538	10538	9298	9298
Rating dummies	Yes	Yes	Yes	Yes
Rating dummies × Cash flow	Yes	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes	Yes

(a) Table 5. Analyst forecast dispersion

(b) Table 6. Investment-cash flow sensitivity

Read: Table 5 reports lower analyst forecast dispersion for firms with financial expert CEOs in cross-sectional regressions, though not in firm fixed effects. Table 6 shows that the interaction of financial expertise and cash flow is negative and significant across specifications.

Economic message: The negative investment-cash flow interaction is one of the paper's strongest mechanism results: financial expert CEOs appear to reduce dependence on internally generated funds, consistent with better access to external finance.

7.4 Continuation of investment-cash flow table and shock responses

Table 6 (continued)

	(1)	(2)	(3)	(4)
Year dummies × Cash flow	Yes	Yes	Yes	Yes
Industry dummies	Yes	No	Yes	No
Industry dummies × Cash flow	Yes	Yes	Yes	Yes
Firm fixed effects	No	Yes	No	Yes
Adjusted R-squared	0.515	0.722	0.733	0.781

(a) Table 6 continued

Table 7

The dependent variable in columns 1 and 2 is the log of cash to assets. The dependent variable in columns 3 and 4 is a dummy variable equal to one if the firm issues debt in a given year. The dependent variable in columns 5 and 6 is the sum of dividends and share repurchases normalized by net income. All regressions include industry and year dummies. Variable definitions are as defined in Table 1. Standard errors clustered at firm-level in parentheses. ***p < 0.01, **p < 0.05, *p < 0.1.

	(1)	(2)	(3)	(4)	(5)	(6)
	Cash holdings		Debt issues		Total payout	
Financial expert CEO × Default spread	0.267** (0.128)	0.218* (0.112)	0.074* (0.041)	0.051 (0.044)		
Financial expert CEO × Post-2003					0.105** (0.046)	0.092* (0.054)
Financial expert CEO	-0.345*** (0.114)	-0.295*** (0.109)	-0.051 (0.035)	-0.002 (0.042)	0.037 (0.028)	0.079 (0.049)
Age	-0.029 (0.022)	-0.051* (0.026)	0.006 (0.007)	-0.009 (0.010)	0.010 (0.013)	-0.021 (0.026)
Age squared	0.000 (0.000)	0.000** (0.000)	-0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)	0.000 (0.000)
Sex	-0.336* (0.199)	-0.078 (0.387)	0.045 (0.039)	-0.026 (0.061)	-0.003 (0.302)	-0.048 (0.191)
Tenure	0.002 (0.003)	-0.008** (0.004)	-0.000 (0.001)	-0.000 (0.002)	-0.002 (0.002)	0.009* (0.005)
Log(assets)	-0.215*** (0.017)	-0.474*** (0.035)	0.067*** (0.004)	0.102*** (0.014)	0.042*** (0.010)	0.077** (0.036)
Log q	0.622*** (0.035)	0.459*** (0.036)				
Leverage	0.078 (0.082)	0.305*** (0.078)			-0.115 (0.081)	0.433*** (0.158)
Cash flow	1.040*** (0.100)	0.466*** (0.084)				
Asset volatility	1.240*** (0.107)	0.411*** (0.087)	-0.218*** (0.034)	-0.032 (0.038)		
Net working capital	-1.148*** (0.131)	-0.494*** (0.133)				
R&D	2.638*** (0.207)	0.610** (0.244)				
Capex	-4.686*** (0.414)	-3.265*** (0.350)	0.940*** (0.108)	0.840*** (0.147)		
Acquisitions	-2.348*** (0.158)	-1.074*** (0.131)				
Dividend dummy	-0.141*** (0.048)	0.100* (0.053)				
MBA		0.016 (0.054)		-0.009 (0.018)		0.048 (0.051)
Econ		-0.051 (0.054)		0.009 (0.018)		-0.088* (0.047)
Science		-0.093 (0.059)		0.015 (0.021)		-0.101 (0.062)
Law		-0.010 (0.083)		0.031 (0.029)		-0.027 (0.095)
Ivy League alumnus		-0.014 (0.063)		-0.007 (0.022)		-0.011 (0.059)
Generalist index		0.035 (0.028)		-0.005 (0.009)		0.014 (0.025)
Recession graduate		0.007 (0.050)		0.009 (0.018)		0.036 (0.049)
Fast track CEO		-0.003 (0.003)		0.001 (0.001)		0.006e (0.004)
Asset growth					-0.344*** (0.041)	-0.340*** (0.052)
Obs	16191	13594	16295	13682	13184	11169
Industry dummies	Yes	No	Yes	No	Yes	No
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	No	Yes	No	Yes	No	Yes
Adjusted R-squared	0.552	0.789	0.246	0.436	0.069	0.037

(b) Table 7. Credit-market and tax-cut responses

Read: The continuation of Table 6 reports fixed effects and controls for the investment regressions. Table 7 shows that financial expert CEOs increase cash more when default spreads are high, have some evidence of higher debt issuance under tight credit conditions, and increase total payout more after the 2003 tax cuts.

Economic message: Financial expertise matters most when financial decisions are more complex. Tight credit markets and shareholder tax changes are settings where the payoff to financial sophistication should be high.

7.5 CFO changes, lender links, and CEO-firm matching

Table 8

Column 1 shows the effect of financial experience on CFO turnover. The dependent variable in this regression is a dummy variable which is equal to one if the CFO is replaced in the first year of tenure of a new CEO. CEO and firm controls are CEO age, CEO age squared, sex, tenure, log assets, log q, book leverage, cash flow, asset volatility, net working capital, R&D, capex, acquisition dummy, and dividend dummy. Columns 2 and 3 show the effect of financial experience on choice of lending institutions. The dependent variable in these regressions is a dummy variable that is equal to one if the CEO is obtaining the first loan facility after appointment from a bank the company was not doing business with before. In column 2 only the last loan facility before the appointment is considered. In column 3 we consider all loan facilities in the last three years before the CEO was appointed. Variable definitions are as defined in the Appendix. Standard errors clustered at firm-level in parentheses. ***p < 0.01, **p < 0.05, *p < 0.1.

	(1)	(2)	(3)
	CFO turnover	Switch banks	
	CFO is replaced within the first year after CEO turnover	Different from last bank before CEO turnover	Different from 3 years before CEO turnover
Financial expert CEO	0.070** (0.034)	0.015 (0.049)	-0.051 (0.049)
Age	-0.007 (0.028)	-0.032 (0.036)	-0.045 (0.032)
Age squared	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Sex	-0.165* (0.088)	-0.329* (0.134)	-0.210 (0.137)
Tenure	0.009 (0.025)	0.090*** (0.026)	0.047* (0.028)
Log(assets)	0.016 (0.013)	-0.035* (0.018)	-0.048** (0.019)
Log q	-0.042 (0.035)	-0.100* (0.060)	-0.091 (0.058)
Leverage	0.008 (0.116)	-0.054 (0.175)	-0.244 (0.175)
Cash flow	-0.195** (0.096)	0.123 (0.135)	0.149 (0.128)
Asset volatility	0.940 (0.124)	0.198 (0.216)	0.122 (0.210)
Net working capital	0.099 (0.103)	-0.260 (0.178)	-0.372*** (0.173)
R&D	-0.153 (0.188)	0.488 (0.541)	0.726 (0.597)
Capex	-0.189 (0.409)	-1.628** (0.693)	-0.685 (0.743)
Acquisitions	-0.180 (0.226)	-0.141 (0.377)	-1.017*** (0.350)
Dividend dummy	-0.022 (0.037)	0.008 (0.055)	-0.032 (0.055)
Log(cash)	0.027 (0.013)	0.003 (0.020)	-0.007 (0.020)
MBA	-0.027 (0.033)	-0.089* (0.051)	-0.058 (0.053)
Econ	-0.017 (0.035)	-0.057 (0.050)	-0.053 (0.051)
Science	-0.028 (0.038)	-0.071 (0.058)	-0.362*** (0.057)
Law	-0.141** (0.055)	-0.113 (0.090)	-0.082 (0.086)
Ivy League alumnus	-0.072* (0.042)	0.062 (0.058)	0.001 (0.059)
Generalist index	0.051*** (0.019)	-0.003 (0.026)	0.008 (0.026)
Recession graduate	-0.007 (0.034)	0.042 (0.048)	0.014 (0.053)
Fast track CEO	0.004 (0.004)	0.008* (0.005)	0.004 (0.005)
Obs	1017	936	936
Industry dummies	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes
R-squared	0.114	0.279	0.286

(a) Table 8. CFO turnover and bank switching

Table 9

Specifications 1-10 analyze turnover only and the sample is restricted to firm-year observations of turnover years, i.e., includes only firm-year before replacement of CEO in the next year. Specifications 11 and 12 analyze a subsample of these turnovers. Only firms with at least two different turnovers (i.e., the firm is hiring a financial expert in one and a nonfinancial expert in the other turnover) are included. The dependent variable in regressions 1-12 is a dummy equal to one if the incoming CEO is a financial expert. Specifications 13 and 14 analyze all firm years. The dependent variable is a dummy variable that is equal to one if the current CEO is a financial expert CEO. Standard errors are clustered at firm level. Variable definitions are as defined in Table 1. Standard errors clustered at firm level in parentheses. ***p < 0.01, **p < 0.05, *p < 0.1.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
	Incoming CEO is financial expert: conditional of a turnover in t-1													
Log(assets)	0.045*** (0.007)				0.039*** (0.006)	0.034*** (0.011)			0.008 (0.014)			0.001 (0.022)	0.047*** (0.003)	0.047*** (0.003)
Capex		-0.462*** (0.133)			-0.345*** (0.100)	-0.309** (0.136)			-0.392 (0.216)			-0.383 (0.246)	-0.369*** (0.054)	-0.403*** (0.062)
Log q			-0.019 (0.018)			0.053 (0.021)			0.017 (0.029)			0.012 (0.036)	0.005 (0.052)	0.009 (0.050)
Asset volatility				-0.206*** (0.056)		-0.031 (0.025)			-0.138 (0.196)			-0.111 (0.272)	-0.190*** (0.052)	-0.098*** (0.050)
Firm age					0.003*** (0.001)	0.001 (0.001)			0.000 (0.001)			0.001 (0.002)	0.002*** (0.000)	0.002*** (0.000)
Asset growth					-0.064 (0.039)	-0.097** (0.040)			-0.180*** (0.060)			-0.215* (0.112)	-0.032** (0.013)	-0.033** (0.014)
Retained earnings					0.005 (0.007)	-0.006 (0.007)			-0.003 (0.009)			-0.008 (0.009)	-0.003 (0.003)	-0.003 (0.003)
Obs	1871	1871	1871	1816	1858	1737	1865	1722	1752	600	600	1758	1738	1738
Sample	Turnover	Turnover	Turnover	Turnover	Turnover	Turnover	Turnover	Turnover	Turnover	Turnover (non different event)	Turnover (non different event)	Full	Full	Full
Industry × Year dummies	No	No	No	No	No	No	No	No	Yes	No	Yes	Yes	No	Yes
R-squared	0.023	0.005	0.001	0.007	0.011	0.001	0.000	0.020	0.313	0.020		0.463	0.049	0.307

(b) Table 9. CEO-firm matching

Read: Table 8 shows that incoming financial expert CEOs are more likely to replace the CFO in the first year, but there is little evidence that direct prior employer bank links drive the results. Table 9 shows that financial expert CEOs are more likely to be matched with mature firms: larger, older, lower-growth, less volatile firms.

Economic message: Financial expertise appears to affect the internal finance function, but selection is real. The matching table is important because it motivates the paper's caution about causal interpretation.

7.6 Life-cycle controls and retirement-turnover tests

Table 10

In these regressions we explicitly control for the life cycle of the firm by including size, asset volatility, firm age, total investment, asset growth, retained earnings, and by including industry $\times$ year dummies accounting for the fact that the life cycle of the industry is a proxy for the life cycle of the firm. The dependent variable in regression (1) is the log of cash and marketable securities. The dependent variable in regression (2) is the ratio of total debt to assets. The dependent variable in regression (3) is a dummy variable equal to one if the firm paid a dividend in the current year. The dependent variable in regression (4) is a dummy variable equal to 1 if the firm repurchased shares in the current year. Year dummies are included in all regressions. Variable definitions are as defined in Table 1. Standard errors clustered at firm-level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	(1) Cash	(2) Leverage	(3) Dividends	(4) Repurchases
Financial expert CEO	-0.104** (0.050)	0.015** (0.006)	0.004 (0.009)	0.037* (0.019)
Age	-0.006** (0.003)	0.003 (0.006)	0.003 (0.011)	-0.002 (0.011)
Age squared	0.001** (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
Sex	-0.211 (0.161)	-0.007 (0.017)	-0.025 (0.041)	-0.060 (0.055)
Tenure	-0.004 (0.004)	0.001 (0.001)	0.002 (0.002)	0.002 (0.002)
Log(assets)	-0.567*** (0.041)	0.012* (0.006)	0.032*** (0.006)	0.050*** (0.010)
Log q	0.389*** (0.042)	-0.009 (0.007)	0.021*** (0.007)	-0.072*** (0.022)
Asset volatility	0.268*** (0.094)	-0.172*** (0.016)	-0.043** (0.022)	-0.131*** (0.051)
Dividend dummy	0.142** (0.056)	-0.022** (0.008)		
ROA		-0.241*** (0.039)	0.055 (0.044)	0.724*** (0.308)
Leverage	0.323*** (0.084)		-0.093*** (0.027)	-0.272*** (0.061)
Cash flow	0.285*** (0.083)			
Net working capital	-0.401*** (0.142)			
Acquisitions	-1.306*** (0.148)			
Dividend dummy ($t-1$)			0.593*** (0.019)	0.018 (0.026)
Cash			0.022** (0.009)	0.025* (0.028)
MBA	0.016 (0.055)	0.001 (0.008)	-0.006 (0.010)	0.048** (0.023)
Econ	-0.038 (0.056)	0.006 (0.007)	-0.006 (0.010)	-0.027 (0.023)
Science	-0.077 (0.061)	0.001 (0.008)	-0.002 (0.011)	-0.033 (0.026)
Law	-0.077 (0.084)	-0.015 (0.010)	-0.029 (0.018)	0.004 (0.045)
Ivy League alumnus	-0.011 (0.063)	0.006 (0.008)	-0.013 (0.010)	-0.031 (0.025)
Generalist index	0.051* (0.027)	-0.000 (0.003)	-0.003 (0.005)	0.011 (0.012)
Recession graduate	-0.010 (0.053)	-0.000 (0.006)	-0.006 (0.009)	0.007 (0.022)
Fast track CEO	-0.002 (0.003)	0.000 (0.000)	0.001 (0.001)	0.001 (0.001)
Firm age	0.008 (0.005)	-0.000 (0.001)	0.002** (0.001)	0.003 (0.002)
Total investments	-2.563*** (0.288)	-0.197*** (0.061)	0.067 (0.048)	0.070 (0.132)
Asset growth	0.168*** (0.037)	-0.039*** (0.006)	-0.078*** (0.008)	-0.078*** (0.019)
Retained earnings	0.006 (0.009)	-0.001 (0.003)	0.001 (0.002)	0.007 (0.004)
Obs	12423	12493	11891	11891
Industry $\times$ Year dummies	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes

(a) Table 10. Controls for firm life cycle

Table 11

while restricting the sample to firms headed by CEOs who are succeeding a CEO who reached retirement age. The specifications are the same as in Table 2. CEO and firm variable are as defined in Table 1. Standard errors clustered at firm-level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	(1) Cash	(2) Leverage	(3) Dividends	(4) Repurchases
Financial expert CEO	-0.205** (0.081)	0.030** (0.013)	0.009 (0.016)	0.049 (0.036)
Age	-0.067** (0.043)	0.005 (0.006)	-0.001 (0.008)	-0.020 (0.019)
Age squared	0.001** (0.000)	-0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Sex	-0.508** (0.253)	-0.028 (0.041)	-0.086 (0.095)	-0.105 (0.120)
Tenure	-0.002 (0.007)	-0.002 (0.001)	-0.000 (0.001)	0.002 (0.003)
Log(assets)	-0.584*** (0.053)		0.023 (0.016)	0.089** (0.041)
Log q	0.375*** (0.065)		-0.014 (0.015)	
Asset volatility		0.135 (0.237)	-0.181*** (0.029)	
R&D		0.708 (0.364)	-0.104 (0.153)	
Capex		-4.017*** (1.007)	-0.342*** (0.112)	
Dividend dummy		0.057 (0.119)	-0.012 (0.017)	
ROA		-0.192** (0.089)	0.026 (0.082)	0.798*** (0.274)
Leverage		0.204 (0.210)		
Cash flow		0.482** (0.207)		
Net working capital		-0.527 (0.361)		
Acquisitions		-1.376*** (0.365)		
Dividend dummy ($t-1$)			0.698*** (0.039)	0.048 (0.067)
Leverage			-0.054 (0.053)	-0.122 (0.137)
Cash			0.004 (0.020)	0.002 (0.054)
MBA	0.063 (0.102)	0.017 (0.102)	0.011 (0.027)	0.037 (0.047)
Econ	-0.011 (0.110)	0.004 (0.110)	-0.000 (0.016)	0.006 (0.044)
Science	0.022 (0.122)	0.019 (0.122)	-0.021 (0.020)	-0.030 (0.058)
Law	0.022 (0.157)	0.009 (0.157)	-0.031 (0.031)	-0.067 (0.093)
Ivy League alumnus	-0.094 (0.099)	-0.094 (0.099)	0.011 (0.025)	-0.063 (0.050)
Generalist index	0.036 (0.057)	-0.013* (0.057)	-0.002 (0.016)	-0.005 (0.024)
Recession graduate	-0.001 (0.014)	-0.001 (0.014)	0.005 (0.015)	0.014 (0.046)
Fast track CEO	0.002 (0.005)	0.000 (0.001)	-0.001 (0.001)	0.001 (0.002)
Firm age	0.014 (0.011)	0.000 (0.001)	0.002 (0.001)	-0.000 (0.003)
Asset growth	0.222*** (0.048)	0.017 (0.048)	-0.002 (0.020)	-0.084* (0.045)
Retained earnings		-0.003 (0.015)	0.002 (0.006)	0.005 (0.012)
Obs	2751	2768	2616	2616
Life cycle controls	Yes	Yes	Yes	Yes
Industry $\times$ Year dummies	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Adjusted R-squared	0.790	0.698	0.893	0.392

(b) Table 11. Turnovers after retiring CEOs

Read: Table 10 adds direct life-cycle controls such as size, age, asset volatility, investment, growth, and retained earnings; the main cash, leverage, and repurchase results remain. Table 11 restricts to turnovers after a CEO reaches retirement age and still finds lower cash, higher leverage, and more payout patterns.

Economic message: These tables are the paper's main attempt to separate financial skills from mature-firm selection. They do not fully prove causality, but they show that the baseline patterns are not mechanically driven by simple life-cycle observables.

7.7 Performance, pay, and matching estimator

Table 12
Columns 1 and 2 show the effect of firm age and financial expertise of the CEO on firm performance. The dependent variable in all regressions is the log of q. Columns 3 and 4 present estimates of OLS and firm fixed effects panel regressions of the logarithm of CEO total and equity pay on the financial expertise of the CEO, and firm-level control variables. CEO and firm variables are as defined in Table 1. Standard errors clustered at firm-level in parentheses.
***p < 0.01, **p < 0.05, *p < 0.1.

	(1)	(2)	(3)	(4)
	Tobin's q		Total pay	Equity pay
Financial expert CEO × Asset growth	-0.122*** (0.040)			
Financial expert CEO × Firm age		0.002** (0.001)		
Financial expert CEO	0.007 (0.017)	-0.056* (0.031)	-0.040 (0.035)	0.002 (0.056)
Firm age	0.002** (0.001)	0.001 (0.001)	0.016*** (0.005)	0.014*** (0.006)
Asset growth	0.374*** (0.027)			
Log(assets)	-0.299*** (0.021)	-0.293*** (0.020)		
Leverage	-0.465*** (0.061)	-0.446*** (0.064)		
PPE	-0.487*** (0.106)	-0.875*** (0.115)		
Capex	1.317*** (0.158)	1.808*** (0.174)		
R&D	0.176 (0.438)	-0.981** (0.490)		
Age	0.003 (0.010)	-0.004 (0.011)	0.028 (0.025)	-0.009 (0.031)
Age squared	-0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)	0.000 (0.000)
Sex	-0.002 (0.051)	-0.017 (0.055)	-0.015 (0.098)	-0.390** (0.189)
Tenure	0.003** (0.001)	0.004*** (0.001)	-0.005 (0.003)	-0.008* (0.005)
Asset volatility			0.423*** (0.086)	0.938*** (0.300)
External hired CEO			0.232*** (0.037)	0.347*** (0.062)
First year CEO			0.157*** (0.035)	0.272*** (0.055)
CEO-chairman dummy			0.020 (0.030)	0.044 (0.047)
Log market value of assets			0.565*** (0.031)	0.776*** (0.041)
Market to book			-0.057*** (0.018)	-0.030 (0.022)
ROA			0.681*** (0.193)	0.090 (0.253)
ROA(-1)			-0.024 (0.167)	0.365 (0.232)
Stock returns			-0.018 (0.021)	-0.262*** (0.030)
Stock returns (-1)			0.094*** (0.017)	0.057** (0.025)
Obs	14782	16317	10929	7423
Year dummies	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Adjusted R-squared	0.730	0.688	0.673	0.686

CEOs, we are able to observe how the two turns of CEOs changes in the tax code (see Sections 4.4 and 4.5) to test

(a) Table 12. Firm performance and CEO pay

Table 13
The dependent variable in column 1 is the log of cash and marketable securities. The dependent variable in column 2 is the ratio of total debt to assets. The dependent variable in column 3 is a dummy variable equal to one if the firm paid a dividend in the current year. The dependent variable in column 4 is a dummy variable equal to one if the firm repurchased shares in the current year. We use the nearest-neighbor estimator (nmatch) from Abadie, Diuker, Ichim, Hox, and Imbens (2004). We use the same set of firm variables as in our baseline regressions (Table 4) for matching. Moreover, we require that the treated and the control observation(s) match exactly on industry (two-digit SIC) and year. Variable definitions are as defined in Table 1. Standard errors in parentheses. ***p < 0.01, **p < 0.05, *p < 0.1.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Cash		Leverage		Dividends		Repurchases	
Financial expert CEO	-0.204*** (0.024)	-0.219*** (0.023)	0.005* (0.003)	0.008*** (0.003)	0.010*** (0.004)	0.014*** (0.005)	0.043*** (0.009)	0.045*** (0.008)
Obs	15,643	15,643	15,720	15,720	14,950	14,950	14,950	14,950
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

(b) Table 13. Matching-estimator results

Read: Table 12 shows that financial expertise is more valuable in mature firms: the interaction with asset growth is negative and the interaction with firm age is positive in Tobin's q regressions. It also shows no evidence that financial expert CEOs are paid at a discount. Table 13 uses a nearest-neighbor matching estimator and finds results similar to the baseline financial policy regressions.

Economic message: Financial expert CEOs seem particularly useful in firms where financial policy is central to value creation. The pay evidence weakens the idea that firms hire financial experts simply because they are cheaper.

7.8 Nonfinancial policies

Table 14

The dependent variable in columns 1 and 2 is R&D spending. The dependent variable in columns 3-5 is the number of patent applications filed by a company in a given year. The dependent variable in columns 6 through 8 is the total number of citations received by the set of patents that a firm applied for in a given year. The dependent variable in columns 9 through 11 is the number of citations (as before) adjusted for time-truncation as in Hall, Jaffe, and Trajtenberg (2001, 2005). We estimate specifications 1 and 2 by OLS. In specifications 3 to 11 we use negative binomial regressions and introduce pre-sample mean scaling as in Aghion, Van Reenen, and Zingales (2013) and Blundell, Griffith, and Van Reenen (1999). Variable definitions are as defined in Table 1. Standard errors in parentheses. ***p < 0.01, **p < 0.05, *p < 0.1.

	R&D		log Patents		log Citations			log Citations (Q)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Financial expert CEO	-0.559*** (0.202)	-0.810*** (0.246)	-0.007 (0.045)	-0.130*** (0.062)	-0.128** (0.071)	-0.029 (0.053)	-0.152*** (0.069)	-0.181*** (0.079)	-0.014 (0.057)	-0.157*** (0.071)	-0.190*** (0.081)
Financial expert CEO × R&D					0.016* (0.009)		0.024*** (0.011)				
Age	0.052 (0.124)	-0.215 (0.141)	0.012 (0.030)	-0.036 (0.036)	-0.010 (0.035)	0.008 (0.033)	-0.040 (0.042)	-0.020 (0.041)	0.005 (0.035)	-0.042 (0.043)	-0.023 (0.042)
Age squared	-0.001 (0.001)	0.001 (0.001)	-0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)	0.000 (0.000)
Gender	0.555 (0.829)	1.714* (0.966)	0.026 (0.148)	0.430** (0.213)	0.262 (0.214)	0.053 (0.197)	0.641*** (0.241)	0.374 (0.263)	0.048 (0.194)	0.444** (0.276)	0.354 (0.271)
Tenure	-0.010 (0.015)	0.008 (0.024)	-0.000 (0.004)	0.001 (0.007)	-0.004 (0.006)	-0.004 (0.005)	-0.004 (0.008)	-0.005 (0.008)	-0.005 (0.005)	-0.005 (0.008)	-0.007 (0.007)
R&D					0.086*** (0.006)		0.088*** (0.008)		0.088*** (0.009)		0.093*** (0.009)
Log(sales)	-0.831*** (0.120)	-1.209*** (0.141)	0.340*** (0.020)	0.119*** (0.028)	0.237*** (0.024)	0.316*** (0.024)	0.086*** (0.032)	0.219*** (0.028)	0.299*** (0.023)	0.071** (0.032)	0.193*** (0.030)
Log(PPE)	0.164 (0.196)	-0.743*** (0.129)	0.197*** (0.047)	-0.010 (0.033)	0.061* (0.032)	0.168*** (0.050)	-0.068* (0.040)	0.009 (0.037)	0.156*** (0.047)	-0.089** (0.042)	-0.009 (0.039)
Stock returns	0.188* (0.101)	0.286** (0.122)	0.065*** (0.020)	0.075*** (0.028)	0.024 (0.023)	0.043* (0.026)	0.056 (0.034)	0.007 (0.031)	0.030 (0.027)	-0.012 (0.036)	-0.012 (0.033)
IOR	-1.397** (0.556)	-0.487 (0.635)	-0.105 (0.149)	0.148 (0.153)	0.312*** (0.145)	-0.031 (0.172)	0.261 (0.176)	0.358** (0.172)	-0.049 (0.174)	0.191 (0.179)	0.324* (0.175)
MBA	0.016 (0.286)	0.136* (0.070)	0.129** (0.066)		0.173** (0.082)	0.161** (0.070)		0.189** (0.084)	0.174** (0.072)		0.174** (0.072)
Econ	-0.004 (0.263)	0.054 (0.075)	0.054 (0.066)	0.077 (0.075)	0.077 (0.066)	0.122* (0.080)	0.091 (0.070)	0.122* (0.082)	0.119* (0.072)	0.079 (0.073)	0.119* (0.072)
Science	0.142 (0.475)	-0.091 (0.153)	-0.197 (0.130)	-0.197 (0.130)	-0.197 (0.130)	-0.197 (0.130)	-0.197 (0.130)	-0.197 (0.130)	-0.197 (0.130)	-0.197 (0.130)	-0.197 (0.130)
Ivy league alumnus	0.775** (0.355)	-0.025 (0.074)	-0.063 (0.064)	-0.063 (0.064)	-0.063 (0.064)	-0.063 (0.064)	-0.063 (0.064)	-0.063 (0.064)	-0.063 (0.064)	-0.063 (0.064)	-0.063 (0.064)
Generalist index	0.132 (0.132)	0.132 (0.132)	0.132 (0.132)	0.132 (0.132)	0.132 (0.132)	0.132 (0.132)	0.132 (0.132)	0.132 (0.132)	0.132 (0.132)	0.132 (0.132)	0.132 (0.132)
Recession graduate	0.242 (0.285)	0.144** (0.065)	0.123** (0.059)	0.144** (0.065)	0.123** (0.059)	0.144** (0.065)	0.123** (0.059)	0.144** (0.065)	0.123** (0.059)	0.144** (0.065)	0.123** (0.059)
Fast track CEO	0.027 (0.019)	0.003 (0.005)	-0.001 (0.006)	-0.001 (0.006)	-0.001 (0.006)	-0.001 (0.006)	-0.001 (0.006)	-0.001 (0.006)	-0.001 (0.006)	-0.001 (0.006)	-0.001 (0.006)
Obs	8042	7601	8042	7601	7601	8042	7601	7601	8042	7601	7601
Industry dummies	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	No
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pre-sample mean scaling	No	Yes	No	Yes	Yes	No	Yes	No	Yes	No	Yes
R squared/Pseudo R squared	0.431	0.249	0.126	0.078	0.122	0.093	0.062	0.086	0.060	0.037	0.054

(a) Table 14. Innovation outcomes

Table 15

The dependent variable in column 1 is the logarithm of advertising spending. The dependent variable in columns 2 and 3 is the logarithm of next year sales. The dependent variable in columns 4 and 5 are total wage expenses divided by the number of employees. Variable definitions are as defined in Table 1. Standard errors in parentheses. ***p < 0.01, **p < 0.05, *p < 0.1.

	Log advertising	Log sales (t + 1)		Average wage		
	(1)	(2)	(3)	(4)	(5)	(6)
Advertising × Financial expertise		-0.048*** (0.015)	-0.010 (0.013)	-0.009 (0.010)		
Log advertising		0.129*** (0.014)	0.119*** (0.012)	0.045** (0.017)		
Financial expert CEO	0.009 (0.078)	0.060 (0.055)	0.028 (0.048)	0.048 (0.042)	-1.190 (1.965)	-0.385 (1.991)
Log(assets)	0.039*** (0.030)	0.761*** (0.038)	0.761*** (0.036)	0.586*** (0.027)	1.623 (1.025)	0.451 (2.430)
Log q	0.205*** (0.068)	0.244*** (0.033)	0.235*** (0.025)	0.2076*** (0.022)	-3.935 (2.647)	-1.547 (1.897)
Firm age	0.005* (0.003)	0.002* (0.001)	0.001 (0.001)	0.005* (0.003)	-0.069 (0.076)	0.127 (0.088)
Cash flow	0.214 (0.209)	0.242*** (0.081)	0.223*** (0.074)	0.167** (0.083)	5.393 (7.624)	0.164 (5.694)
Net working capital	0.016 (0.210)	0.263*** (0.094)	-0.127 (0.104)	0.012 (0.065)	-44.064*** (13.068)	-15.487 (9.558)
Capex	1.531** (0.723)	1.024*** (0.353)	0.397 (0.272)	-0.027 (0.168)	-28.904** (14.366)	2.706 (10.934)
Asset growth	-0.489*** (0.075)	-0.122*** (0.034)	-0.102*** (0.029)	-0.078*** (0.019)	-6.669*** (2.191)	-5.966*** (2.062)
Stock return		0.090*** (0.016)	0.076*** (0.013)	0.032*** (0.011)		0.016 (0.762)
Asset volatility		-0.036 (0.099)	-0.150* (0.066)	-0.152*** (0.050)		7.066 (4.689)
R&D		-1.422*** (0.197)	-1.249*** (0.195)	-0.641** (0.261)		39.633 (57.293)
Retained earnings		0.007 (0.007)	0.004 - 0.004 (0.006)	0.004 (0.005)		0.948 (0.970)
Age		0.017 (0.052)	-0.004 (0.047)	-0.000 (0.040)		-1.481 (2.227)
Age squared		-0.000 (0.000)	0.000 (0.000)	0.000 (0.000)		0.011 (0.011)
Gender		0.005 (0.101)	0.017 (0.101)	0.017 (0.033)		2.716 (3.807)
Tenure		-0.005 (0.004)	-0.004 (0.003)	0.000 (0.002)		0.181 (0.186)
MBA		-0.008 (0.037)	-0.026 (0.026)	-0.022 (0.030)		-3.720* (2.121)
Econ		0.026 (0.038)	0.018 (0.030)	-0.022 (0.025)		0.151 (2.458)
Science		-0.037 (0.040)	-0.006 (0.034)	0.018 (0.024)		2.910 (3.678)
Law		-0.051 (0.079)	-0.035 (0.052)	-0.032 (0.047)		-4.022 (4.183)
Ivy league alumnus		-0.138*** (0.044)	-0.081*** (0.031)	-0.056 (0.036)		1.932 (2.245)
Generalist index		-0.009 (0.016)	-0.009 (0.013)	0.006 (0.012)		0.106 (1.305)
Recession graduate		-0.009 (0.036)	-0.025 (0.029)	0.036 (0.026)		-0.274 (2.153)
Fast track CEO		-0.005 (0.005)	-0.003 (0.000)	-0.000 (0.000)		0.097 (0.097)

(b) Table 15. Advertising, sales, and wages

Read: Table 14 shows that financial expert CEOs are associated with lower R&D spending and fewer patents and citations, although some interactions suggest they may generate more innovation per dollar of R&D. Table 15 finds little evidence that financial expert CEOs systematically reduce advertising or wages.

Economic message: The paper’s results are not simply “financial experts improve everything.” Their advantage is concentrated in financial policy, and there may be an innovation trade-off.

8 Short summary

Financial expert CEOs are CEOs whose prior careers gave them finance-related experience. These CEOs are common, making up about 41% of the sample. Their firms hold less cash, use more leverage, and repurchase shares more often. The paper provides mechanism evidence that they are more financially sophisticated: they avoid the WACC fallacy, reduce investment-cash flow sensitivity, react more actively to credit-market tightness, and respond more to the 2003 payout tax cut.

The main caveat is endogenous matching. Financial expert CEOs tend to be appointed by mature firms, so the results should not be read as a clean experiment. The authors address this using firm fixed effects, life-cycle controls, retirement turnover tests, and matching estimators, but they explicitly do not claim to fully rule out matching.

9 Conclusion

The paper’s central conclusion is that CEO work experience matters for corporate financial policies. Financial expert CEOs appear to bring financial skills that allow firms to operate with lower cash buffers, more debt, and more active payout and financing policies.

The broader economic lesson is that managerial human capital is multidimensional. A finance-oriented CEO can be valuable when firms need access to capital markets or sophisticated financial policy, but that same background may be less useful, or even costly, for innovation-intensive firms. The best interpretation is therefore not that every firm should hire a financial expert CEO, but that financial expertise is especially valuable when the firm’s life-cycle stage makes financial policy central.

The paper also matters methodologically. It shows how researchers can study CEO characteristics while taking CEO-firm matching seriously. The evidence is consistent with financial skills, but the paper’s caution is important: CEO backgrounds, firm life cycles, and corporate policies are jointly determined in the managerial labor market.